

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455736.9639 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.104 +/- 0.036
Transit depth (Rp/Rs)^2	1.08 +/- 0.74 [%]
Orbital Inclination (inc)	84.37 +/- 1.99
Airmass coefficient 1 (a1)	1.0522 +/- 0.0083
Airmass coefficient 2 (a2)	-0.047 +/- 0.074
Scatter in the residuals of the lightcurve fit is	0.79 %
Best Comparison Star	#1 - [390, 115]
Optimal Aperture	4.32
Optimal Annulus	11.24
Transit Duration (day)	0.047 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.104 ± 0.036 vs 0.1326 (deviation 0.79σ)
Duration fit vs published	0.047 d vs 0.0737 d (deviation 1.22σ)
Mid-time O–C	-5.1 min
Depth SNR	2.9
Residual scatter	0.79 %

Light curve

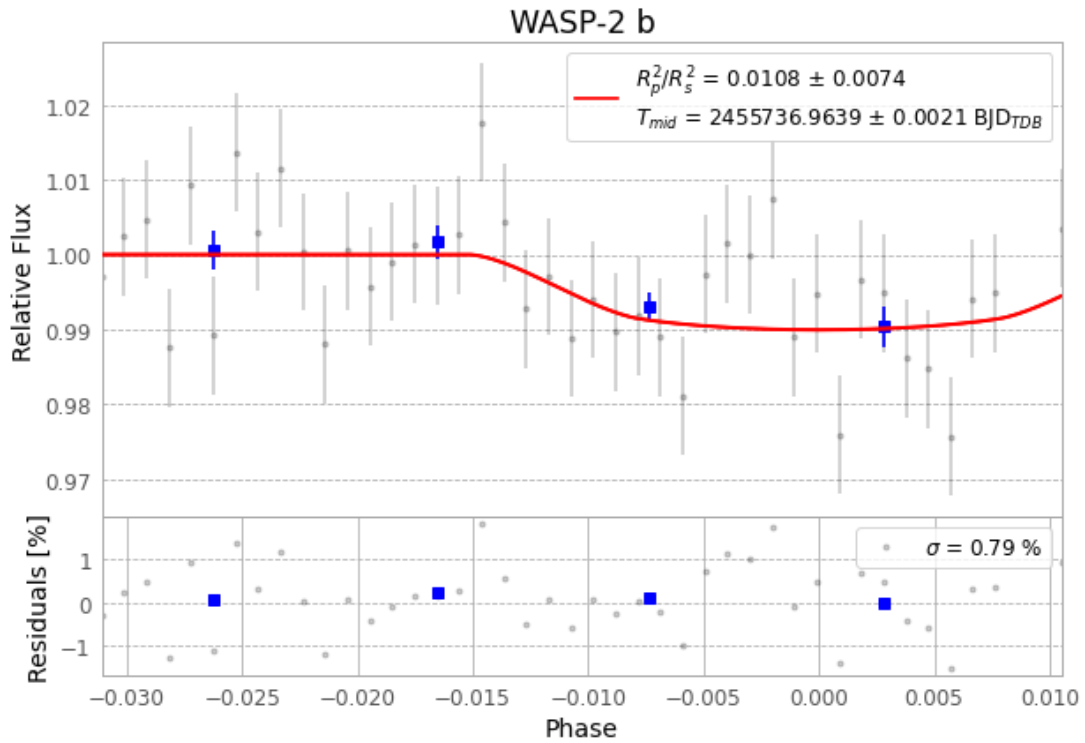


Figure 1 — FinalLightCurve_WASP-2 b_2011-06-24.png (SHA-256 2bcf3f4fe15a3de3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [196, 172], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 93a7331306cff0c5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

52 FITS frames (34 MB), WASP-2110624092413.FITS → WASP-2110624120013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-110624.log (e24cee468c16...), FinalLightCurve_WASP-2 b_2011-06-24.png (2bcf3f4fe15a...), FinalLightCurve_WASP-2 b_2011-06-24.csv (ce51c7deed21...)

Compute resources

Wall time 164.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 23:06Z.